Computer Vision

Assignment 1: Introduction to OpenCV, Responsible Use of Generative AI Tools, and First Steps in Image Processing

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UNIVERSIDAD DE GRANADA



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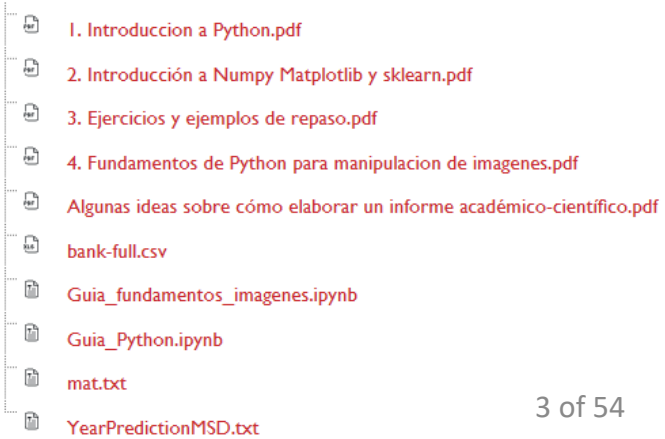
- Submission rules
- Brief introduction to convolutions and filters
- Description and analysis of Assignment 1

Submission Rules

- Only one .ipynb file is submitted, directly **integrating code, results, analysis and discussion**.

- The template/statement is available in PRADO, as well as an example of a correctly solved exercise: Ejemplo_Otsu.ipynb.

- You also have supplementary materials!
  https://pradogrado2627.ugr.es/mod/folder/view.php?id=83733



Submission Rules

- Read very carefully the rules in the template:
  - Upload only the .ipynb file to PRADO. Do not upload images to PRADO!
  - Reading images or any input file: "`/content/drive/MyDrive/images`"
  - Do not write anything into disk (i.e, do not write anything in Google Drive).

Submission

- Deadline: 18 October
  - Late submissions will not be accepted (there are no flexible deadlines).
  - However, the deadline itself is negotiable.
  - My advice is not to ask for an extension on this deadline. The idea is precisely for you to submit it early so that, if possible, we can grade it and give you feedback soon.
- Maximum score: 5 points
- Submission site: PRADO

- **Explanation/discussion/analysis accompanying code and results is essential.**

What is a convolution?

- **Local linear operation** with a mask.
  - The coefficients/values of this mask/filter/kernel determine the operation performed.

If f and g are images, a and b scalar, and L a linear operator:

$$L(af + bg) = aL(f) + bL(g)$$

What is a convolution?

- Technically, a convolution is a *cross-correlation* where the filter/mask is rotated 180 degrees.
  - Unlike correlation, convolution verifies the associative property, which allows us to build complex filters from simple ones.
    - If we have an image f, and we want to convolve it with g and then with h. Knowing that $f * g * h = f * (g * h)$, we can convolve g and h, create a single filter, and then convolve it with f.
  - If the kernel is symmetric:
    - Convolution = Cross-correlation
  - If the kernel is antisymmetric (e.g. [-1 0 1]), only the sign changes

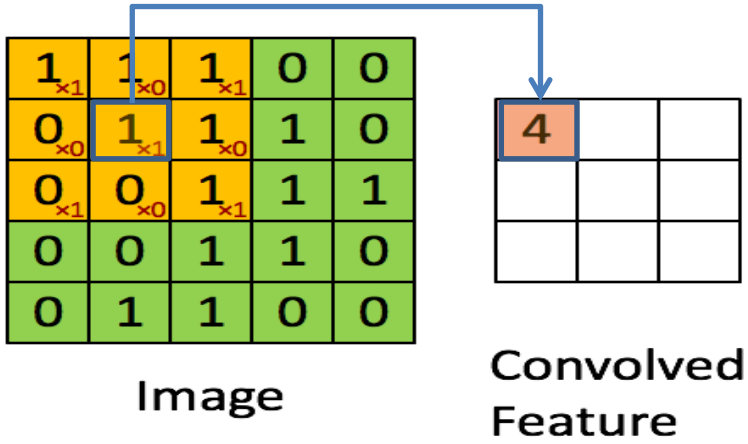
Note: functions like `cv2.filter2D`, actually apply correlation and not convolution.

What is a convolution?

- **Local linear operation** with a mask.

The red numbers represent the values/coefficients of the filter/mask.

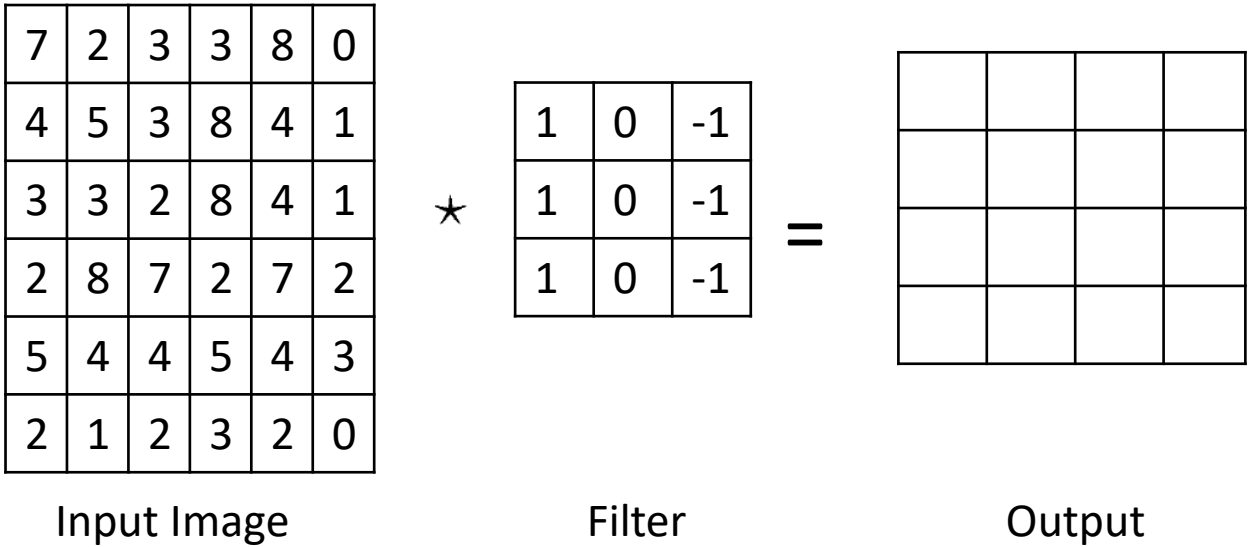
The filter is multiplied element-by-element with the image, the products are added, and the central position of the filter in the image is substituted.



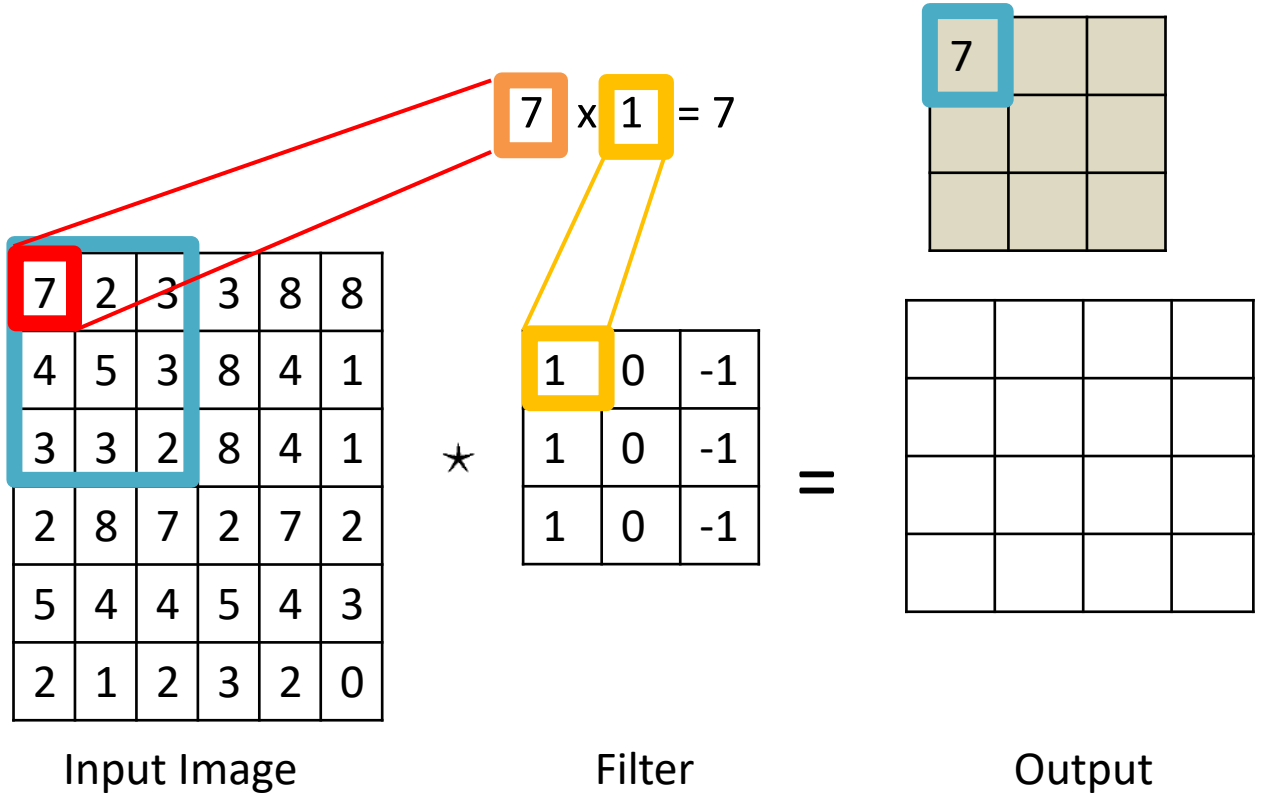


Images extracted from https://glassboxmedicine.com/2019/07/26/convolution-vs-cross-correlation/

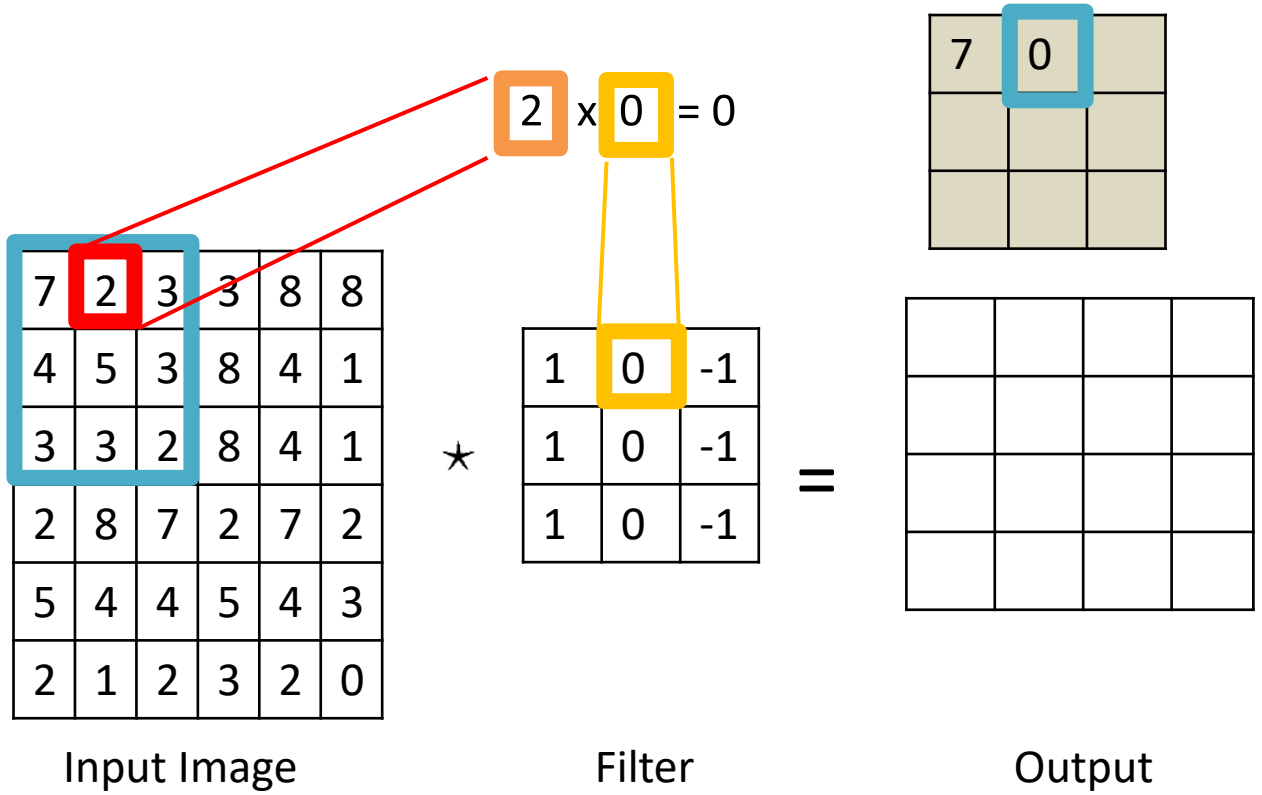
What is a convolution?



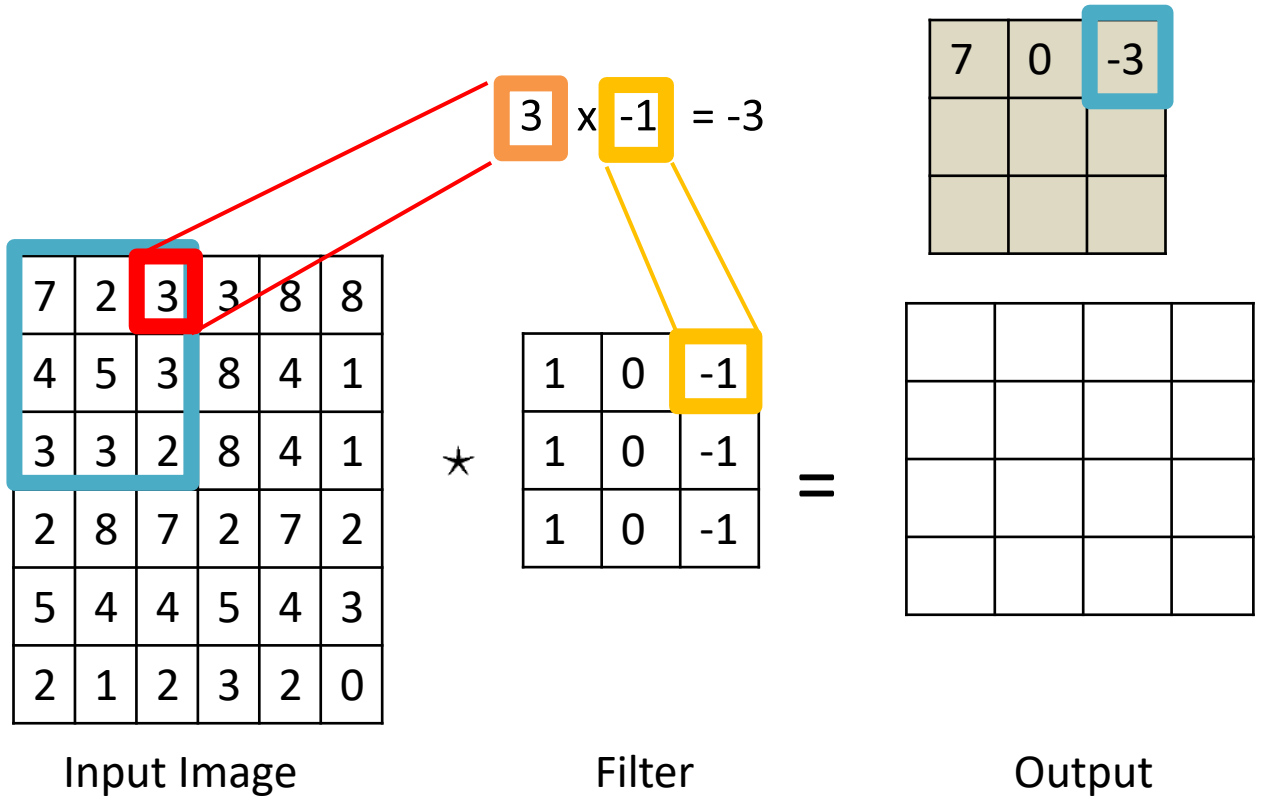
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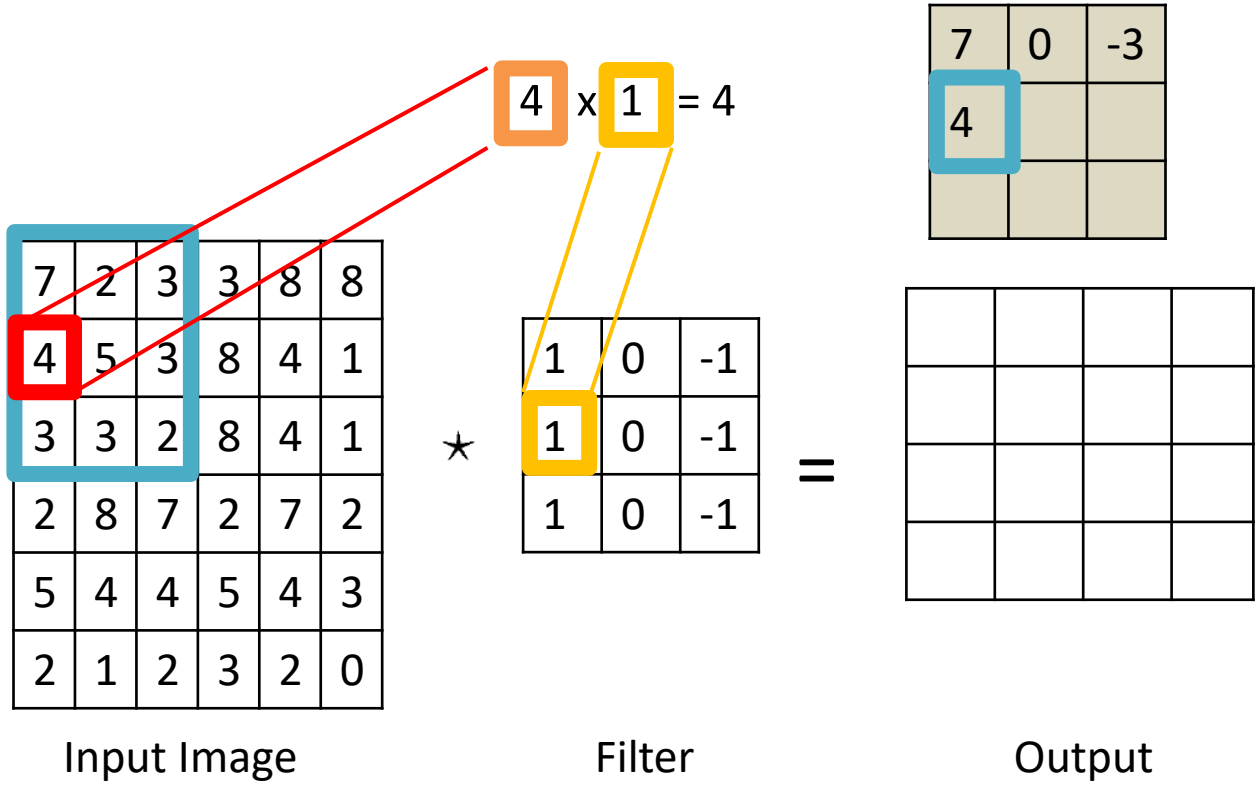
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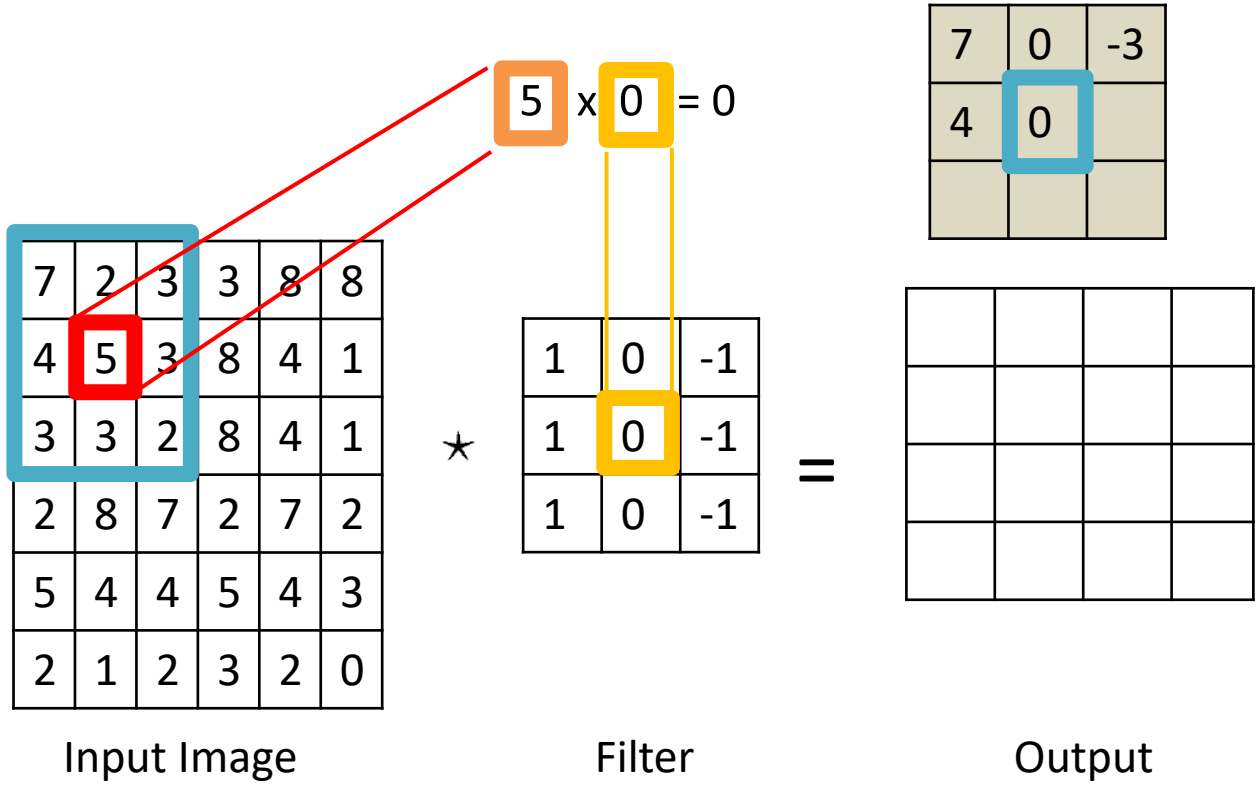
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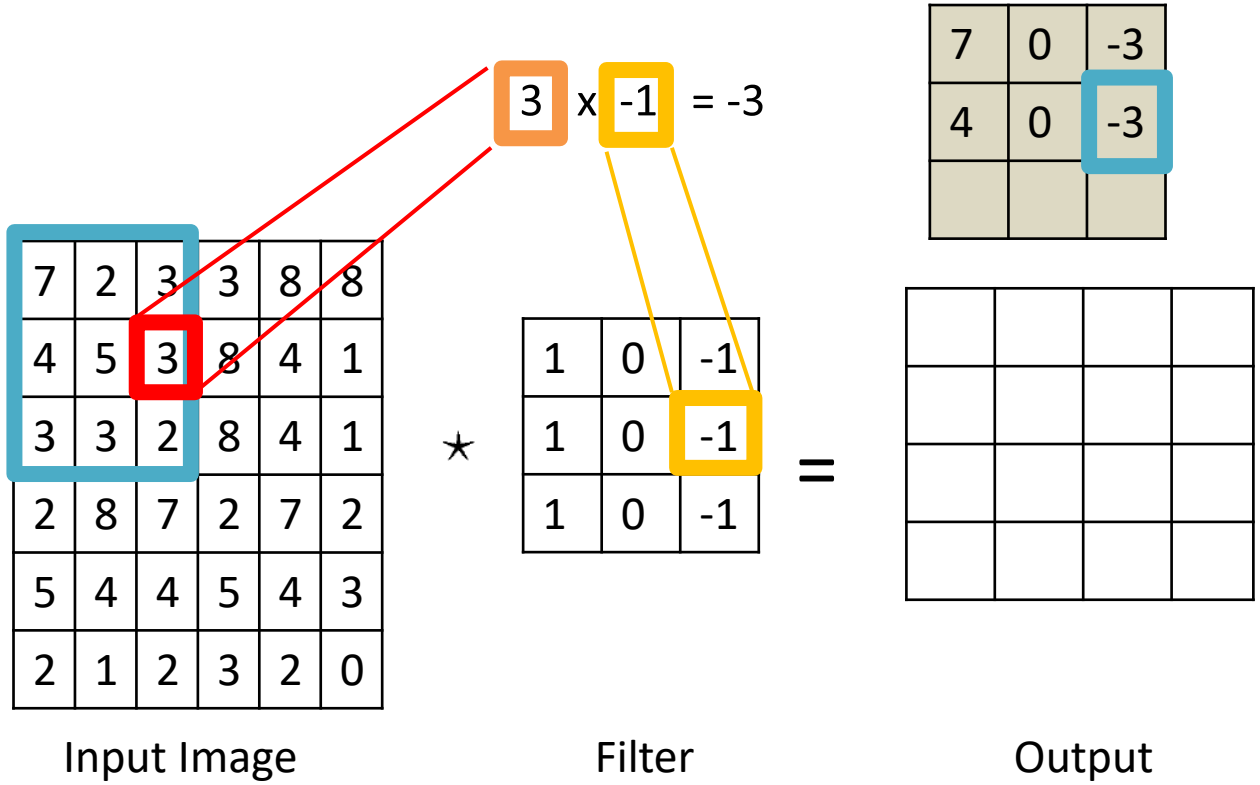
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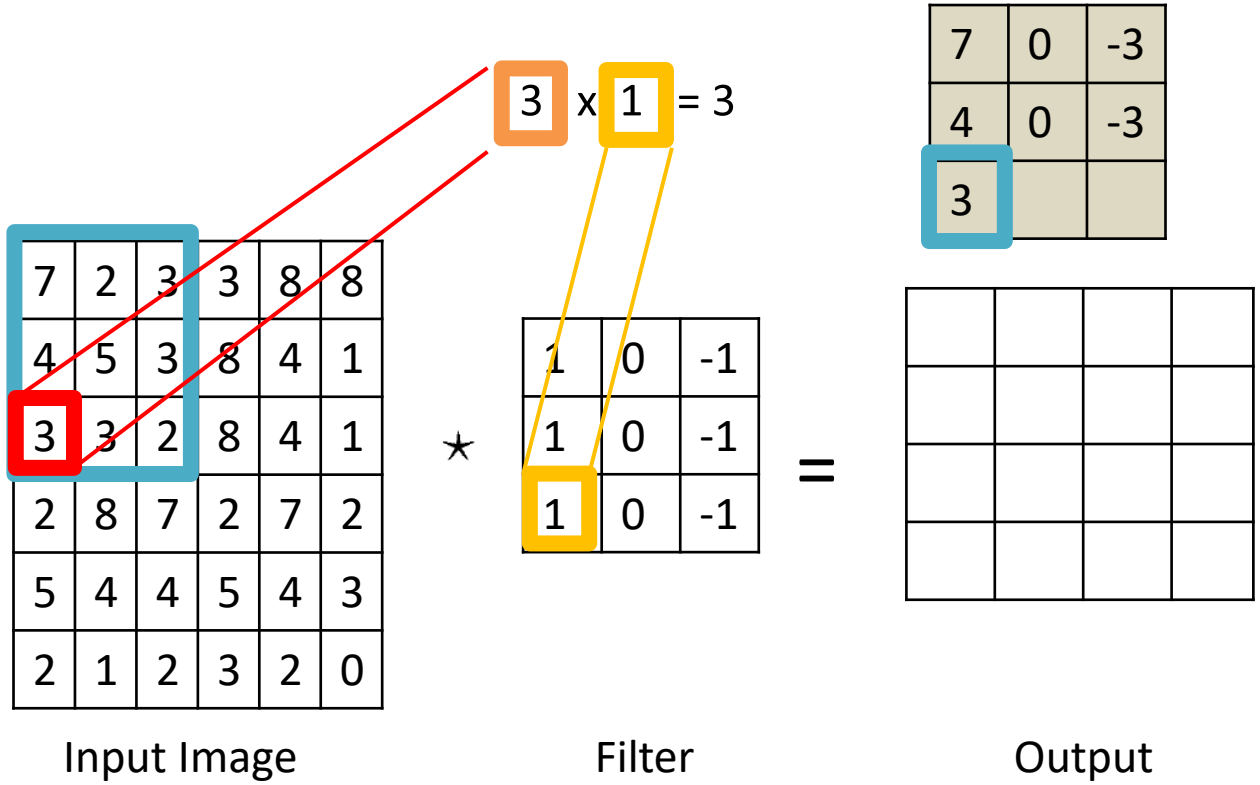
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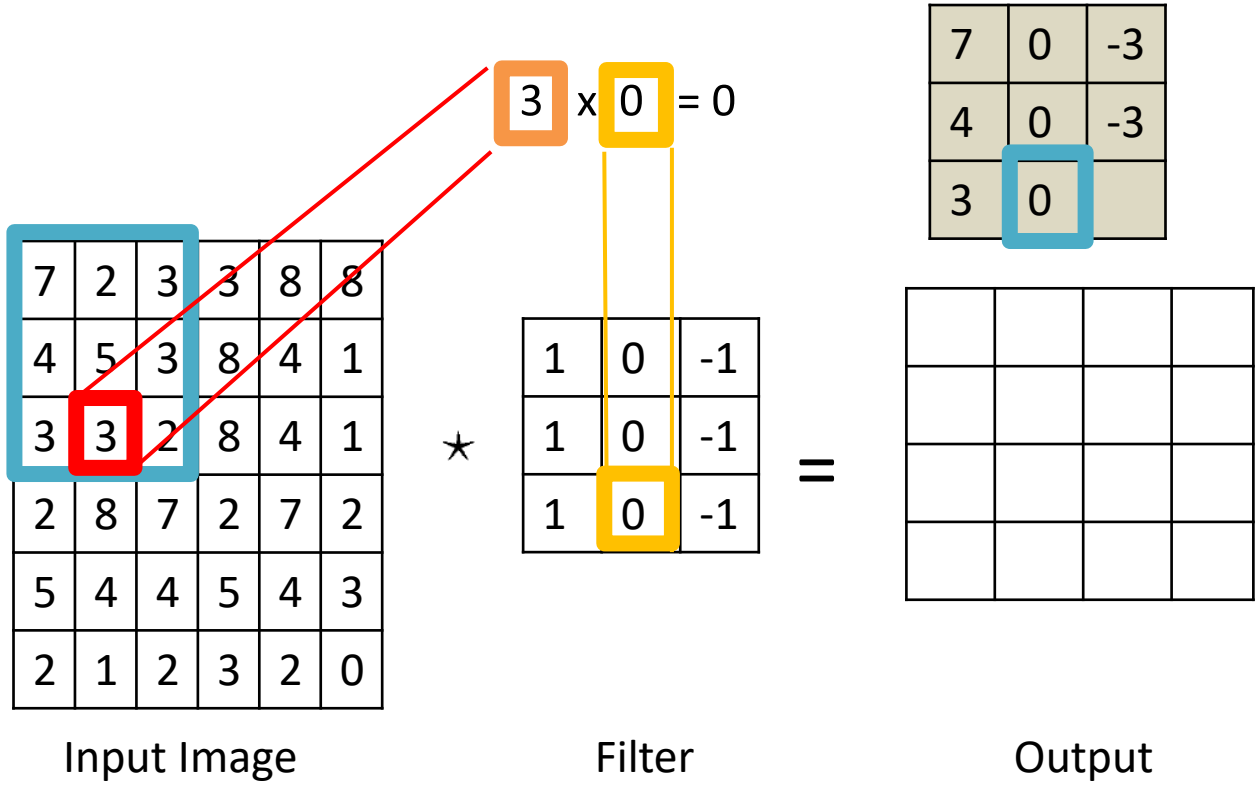
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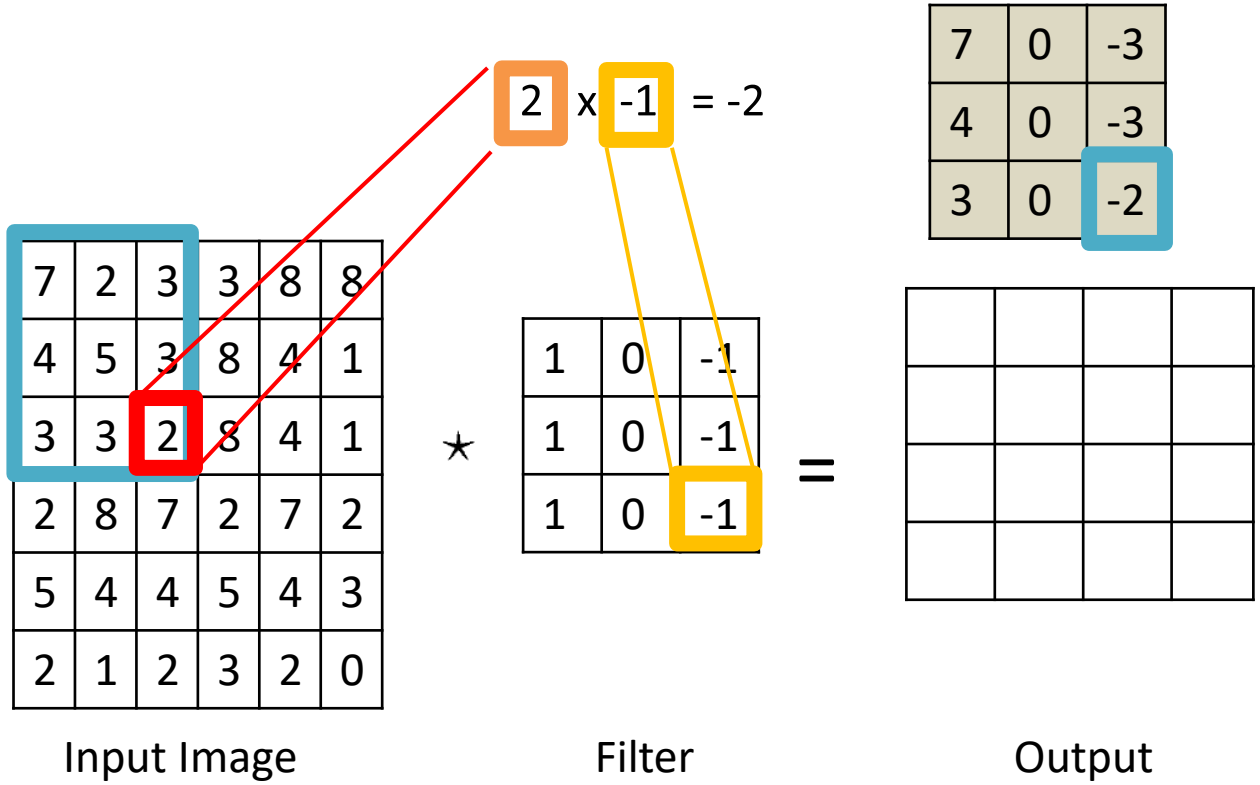
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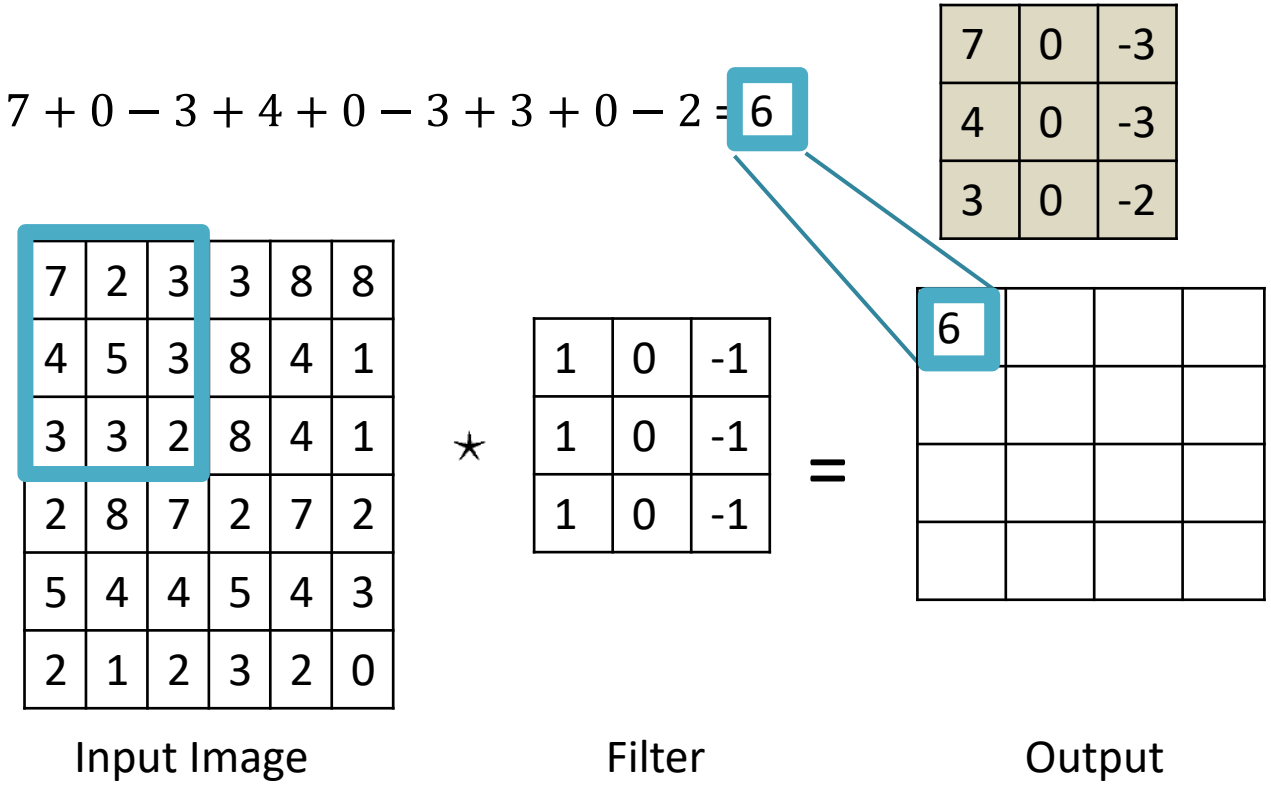
What is a convolution?



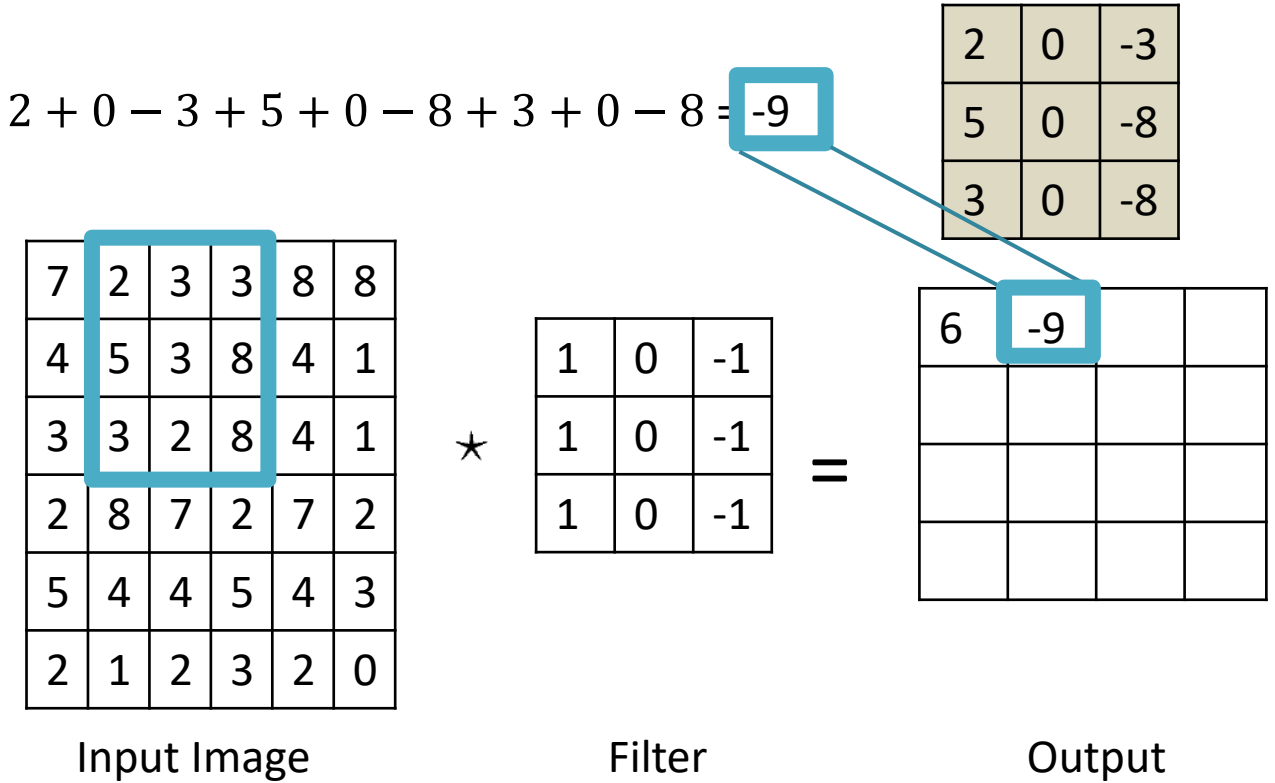
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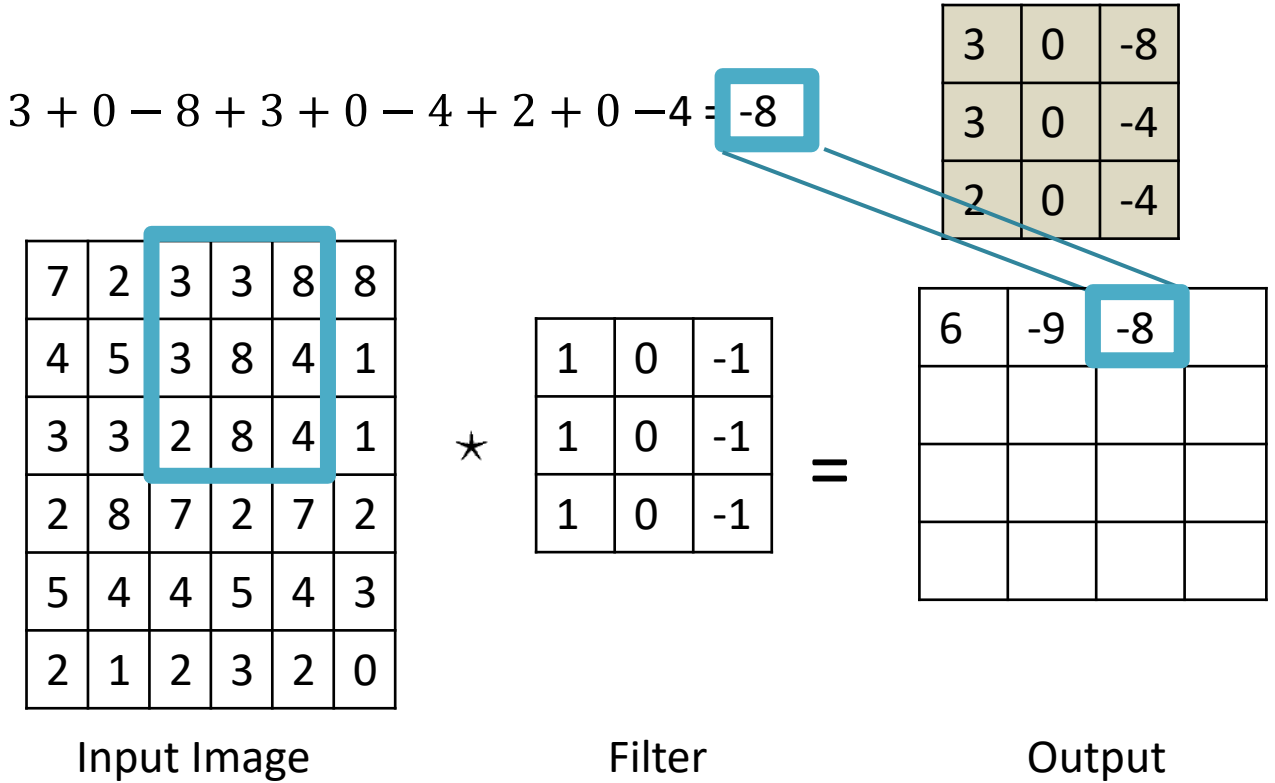
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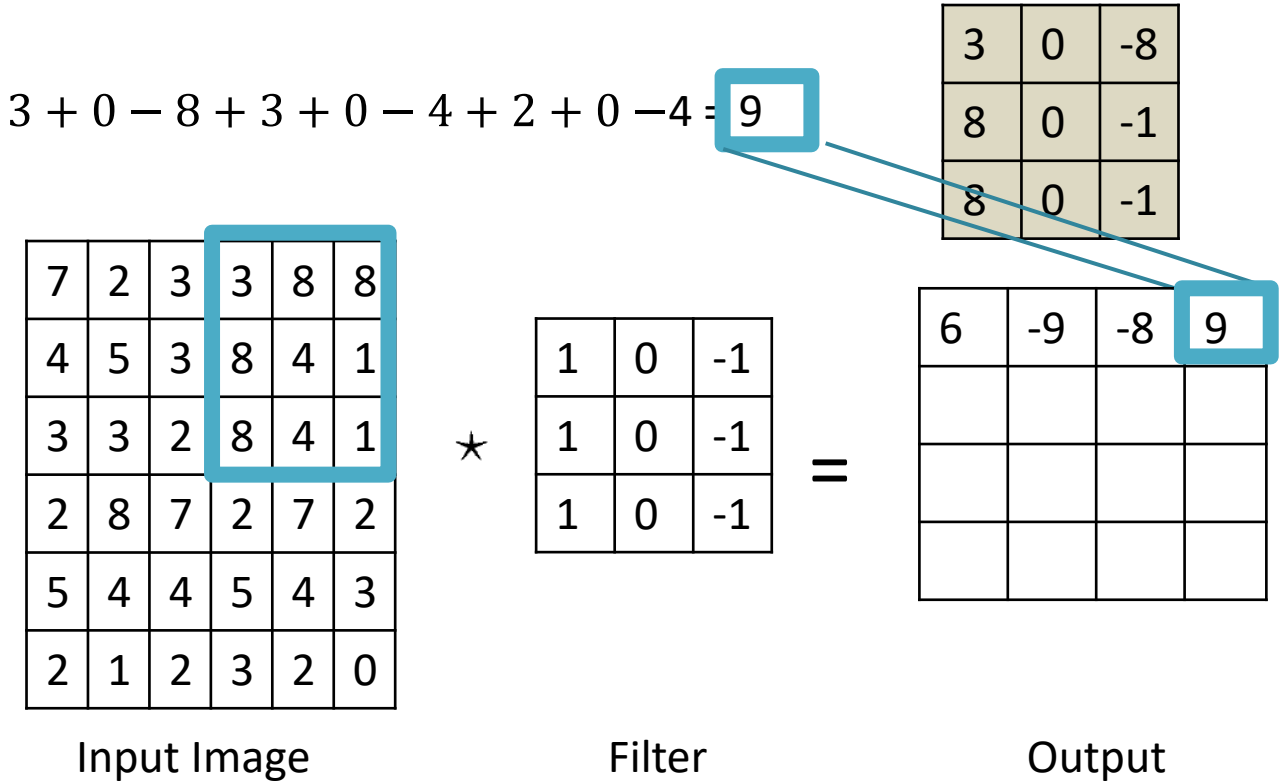
What is a convolution?



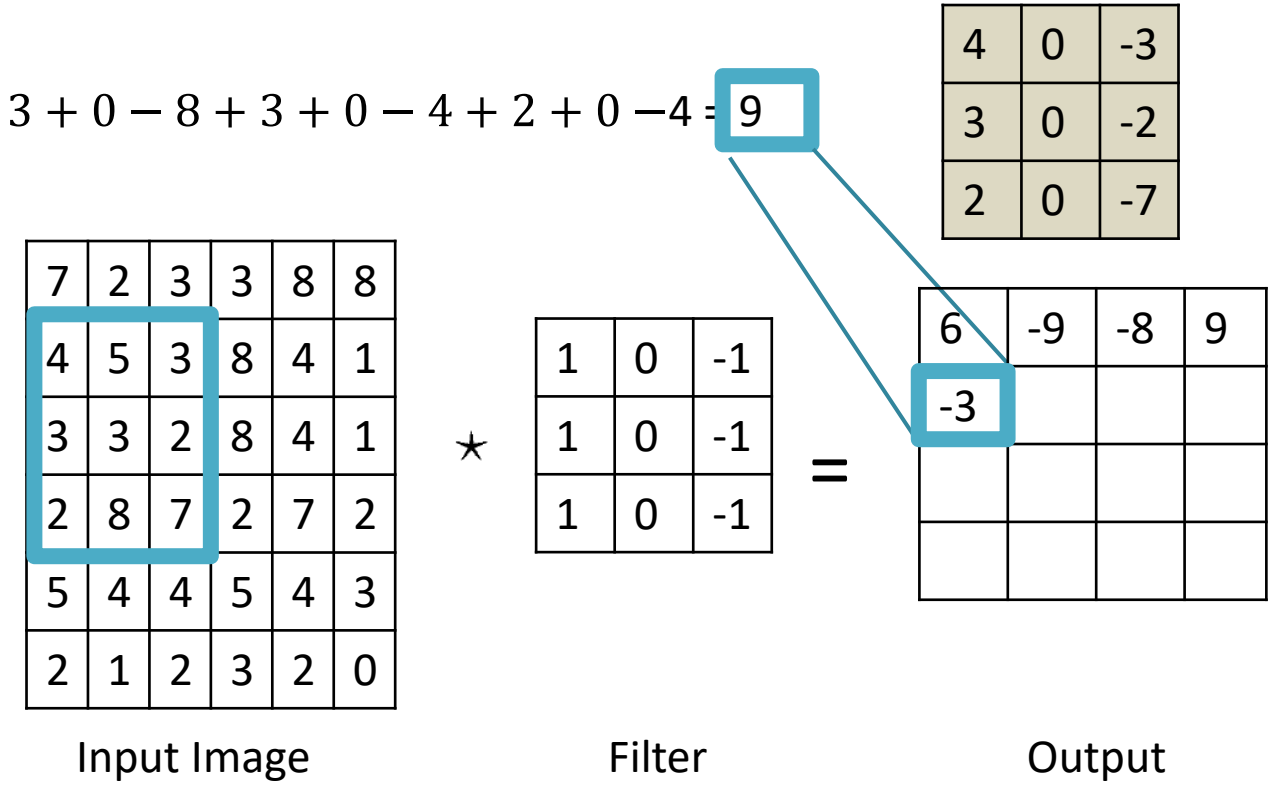
What is a convolution?



What is a convolution?



What is a convolution?



What is a convolution?

- **Local linear operation** with a mask.
  - The values of the mask determine the result (features to be extracted)

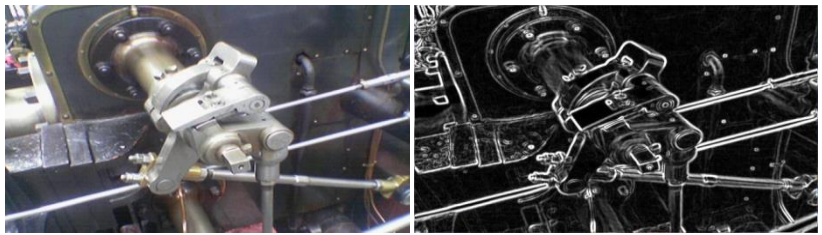
Gaussian filter (attenuates high frequencies → smooths the image)

$$\frac{1}{273}\begin{bmatrix} 1 & 4 & 7 & 4 & 1 \\ 4 & 16 & 26 & 16 & 4 \\ 7 & 26 & 41 & 26 & 7 \\ 4 & 16 & 26 & 16 & 4 \\ 1 & 4 & 7 & 4 & 1 \end{bmatrix}$$

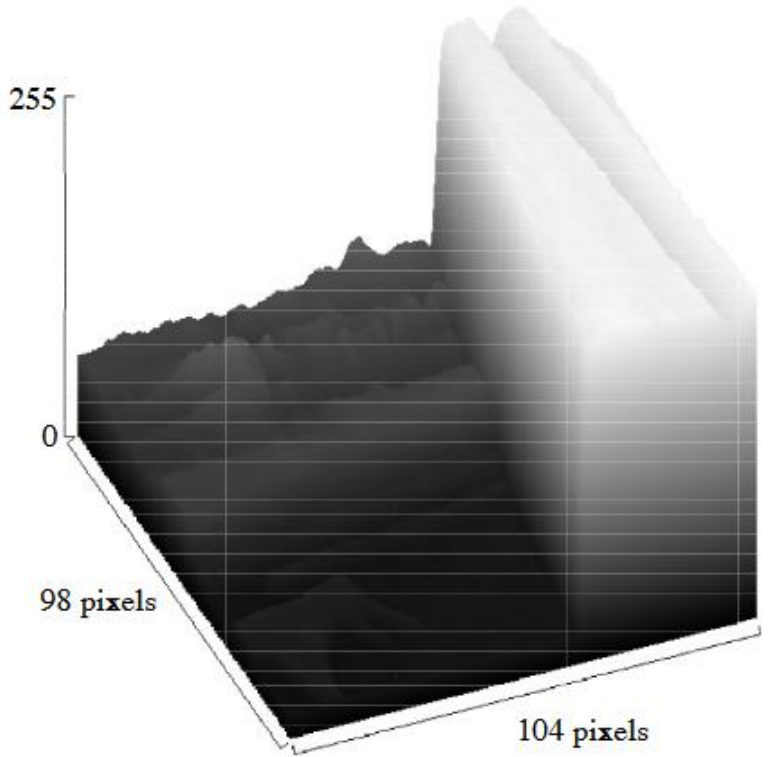


Sobel filter (attenuates low frequencies → enhances edges)

$$\mathbf{G}_x = \begin{bmatrix} +1 & 0 & -1 \\ +2 & 0 & -2 \\ +1 & 0 & -1 \end{bmatrix} \quad \mathbf{G}_y = \begin{bmatrix} +1 & +2 & +1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$



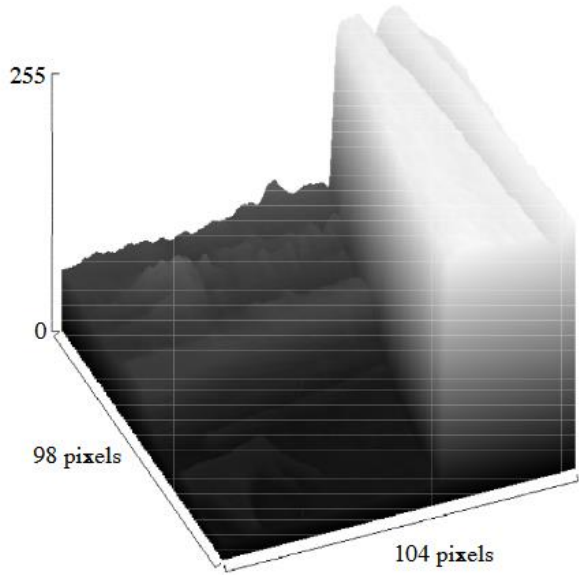
3D visualization of an edge



Example extracted from
https://www.cs.auckland.ac.nz/~rklette/TeachAuckland.html/775/pdfs/D02-Images.pdf

Edges and High Frequencies

- Frequency in 2D images:
  - Rate of change of grey levels with respect to space
    - If it “involves” many pixels to make a change → low frequency
    - If it “involves” few pixels to make a change (i.e., the change is abrupt) → high frequency



The result/effect depends on the kernel







Smoothing/blurring
(to remove noise)

Differentiation
(to highlight details)

Two types of kernels

Smoothing

$$\sum g_i = 1$$

(smoothing a constant function should not change it)

Example: (1/4) * [1 2 1]

Low pass filter (Gaussian)

Derivative/differentiation

$$\sum g_i = 0$$

(differentiating a constant function should return 0)

Example: [-1 0 1]

High pass filter (derivative of Gaussian)

In fact, what happens if you do not divide by the sum of the coefficients?



Advancing ideas…

In ConvNets, these values are learned! They are not defined by a human expert. They are free network parameters and are trained like any other weight.



1989: LeCun et al. used **back-propagation to directly learn the coefficients of convolutional filters** from images of handwritten digits.

1998: LeCun et al. presented **LeNet-5**, a 5-layer ConvNet that could automatically recognise handwritten digits in checks, and **demonstrated strong performance on this task**.

LeCun, Y., Boser, B., Denker, J. S., Henderson, D., Howard, R. E., Hubbard, W., & Jackel, L. D. (1989). Backpropagation applied to handwritten zip code recognition. Neural computation, 1(4), 541-551.

LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). Gradient-based learning applied to document recognition. Proceedings of the IEEE, 86(11), 2278-2324.

Exercise 1

- Focused on:
  - Critical analysis of scientific texts
  - Responsible use of generative AI tools
  - Proper documentation of the work carried out

Exercise 1.A

- **Critical Use of LLMs**: Energy Cost of Generative AI (1 point)
  - Energy consumption associated with both the training and a query or inference performed using generative AI (text, image, video)
  - Conventional literature search (Google Scholar, Scopus, or other academic databases) VS LLM-assisted search

Exercise 1.B

- **Critical Analysis of a Scientific Article** on the Capabilities and Limitations of AI (1 point)
  - Option A: AI Agents and Scientific Research
  - Option B: Multimodal Models and Understanding of the Physical World

Exercise 2.A

- Noise, Filtering, and Edge Detection using **OpenCV functions** (1.5 points)
  - The goal is to start becoming familiar with some of the high-level functions provided by OpenCV.

Exercise 2.A

1. Generation of Noisy Images



Exercise 2.A

2. Image Filtering

Ruido Gaussiano sigma=10

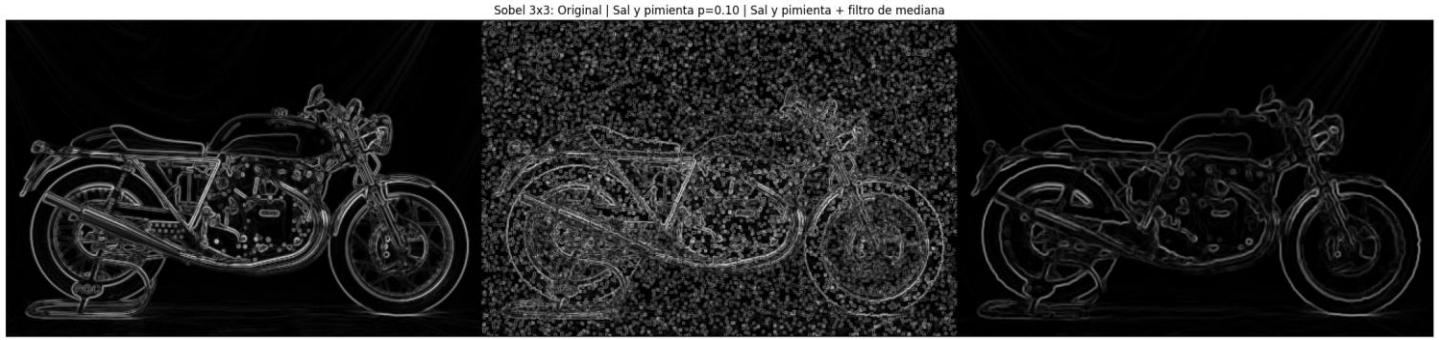


Ruido Gaussiano sigma=10 - kernel 11x11 | Promedio | Gaussiano | Mediana

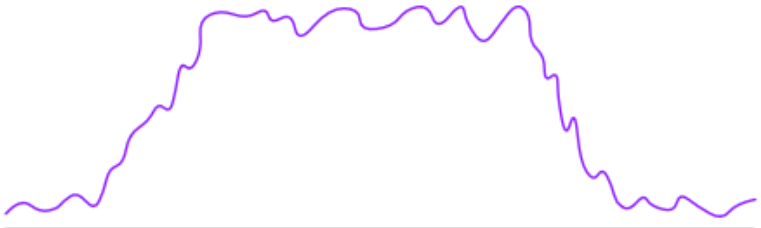


Exercise 2.A

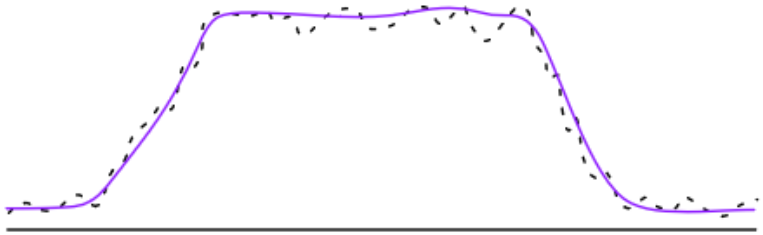
3. Noise, Smoothing, and Edge Detection



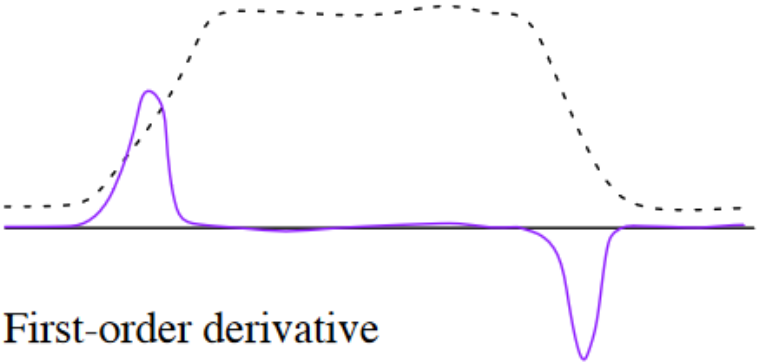
Exercise 2.A



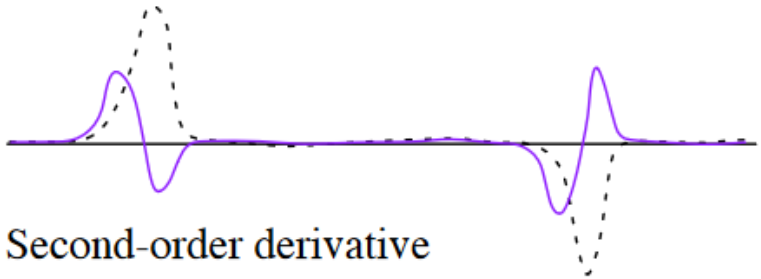
Intensity profile of an input image



After noise removal



First-order derivative



Second-order derivative

Example extracted from https://www.cs.auckland.ac.nz/~rklette/TeachAuckland.html/775/pdfs/D06-LocalOp1.pdf

Exercise 2.A

Sobel filter

Horizontal derivative

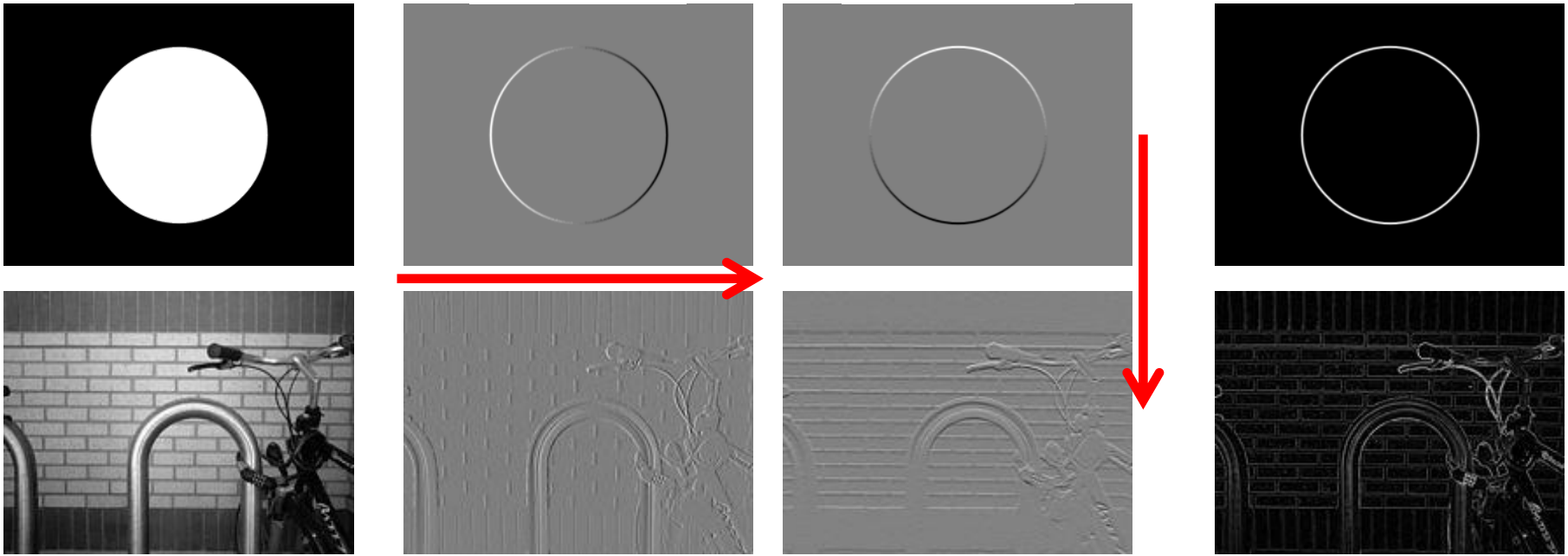
$$\mathbf{G_x} = \begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix}$$

Vertical derivative

$$\mathbf{G_y} = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ +1 & +2 & +1 \end{bmatrix}$$

Gradient magnitude

$$\mathbf{G} = \sqrt{\mathbf{G_x}^2 + \mathbf{G_y}^2}$$



Visual examples taken from https://stackoverflow.com/questions/19815732/what-is-the-gradient-orientation-and-gradient-magnitude and https://en.wikipedia.org/wiki/Sobel_operator

Note: These filters can give negative values, because derivatives can be negative. Then, for visualization purposes, you have to "transfer" those values to [0,255] or [0,1].

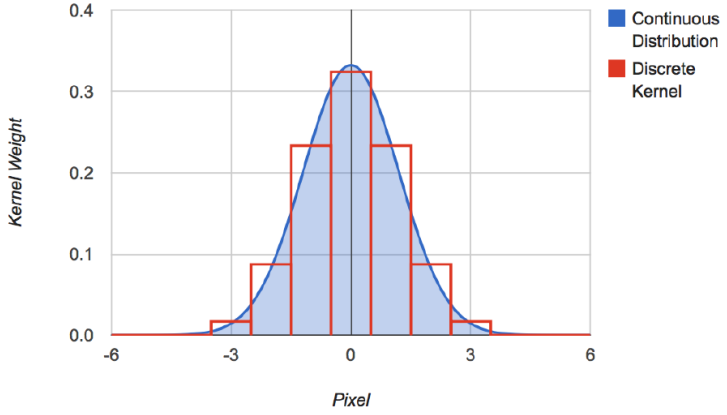
Exercise 2.B

- Calculate the discrete 1D masks of the **Gaussian** function, the **derivative of the Gaussian** and the **second derivative of the Gaussian** (1.5 points)
  - The goal is to start implementing certain image-processing operations at a lower level, avoiding the use of OpenCV high-level functions and relying instead on NumPy functions.

Exercise 2.B

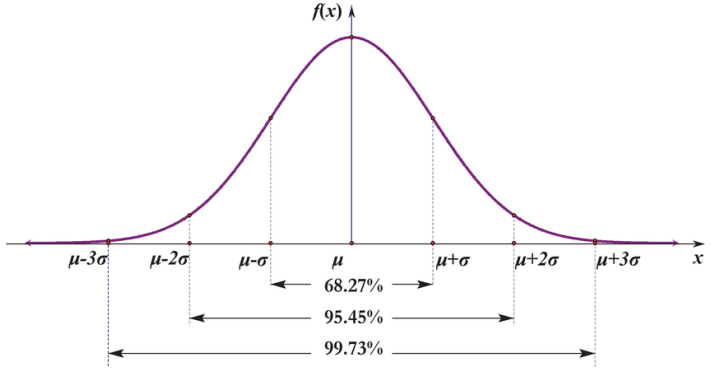
We want to discretize a Gaussian.

There are different approaches to this discretization. Here, we will consider one of them, but students are free to use a different approach, provided that they explain and justify it clearly.



http://dev.theomader.com/gaussian-kernel-calculator/

We know that almost all values are within three standard deviations of the mean → It makes sense to use, by default, 3σ.



Exercise 2.B

$$f(x) = c \cdot e^{-\frac{x^2}{2\sigma^2}}$$

We ignore this normalizing constant ($^1/_{\sigma\sqrt{2\pi}}$)! The shape of the filter is the same! We later normalize the kernel, making the sum of the coefficients equal to 1

$$mask = [f(-k), \dots, f(-1), f(0), f(1), \dots, f(k)], k\ an\ integer$$

We keep the minimum k value that satisfies, $k \geq 3\sigma$

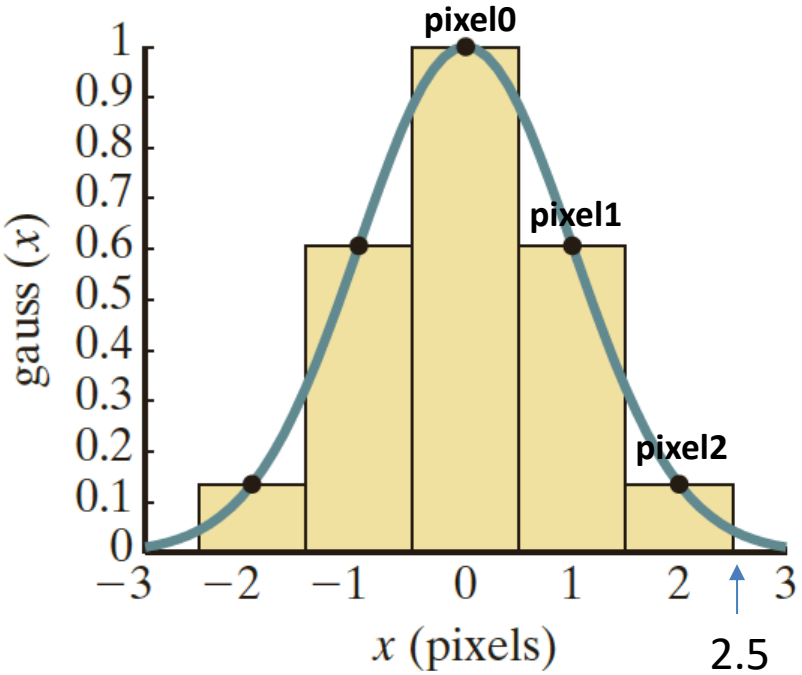
for σ= 1 the mask size is 7 (i.e. $k = 3$).

In addition, $\sum_{i=-k}^{k} f(i) = 1$

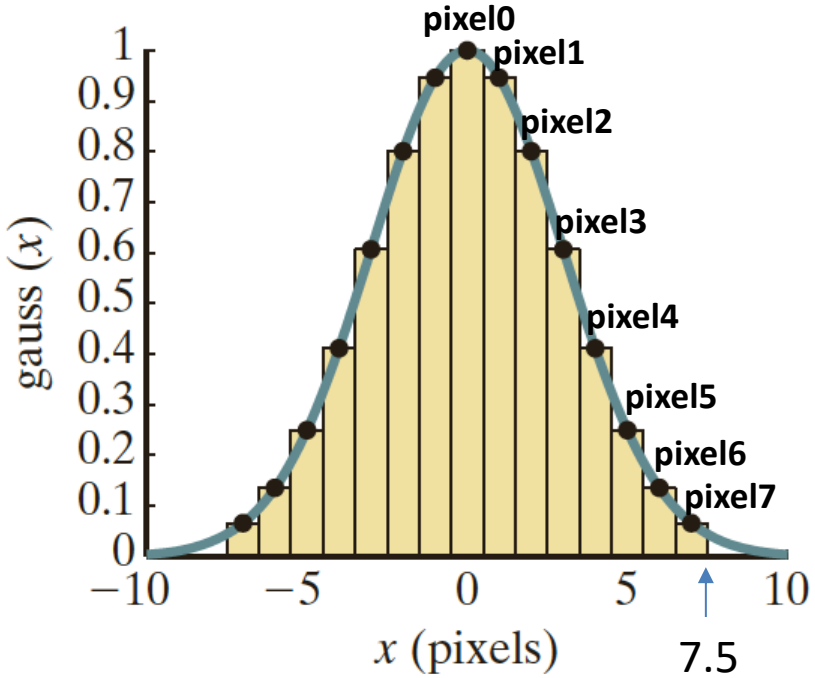
The Gaussian smoothing mask must add up to 1 → remember to divide your mask by the sum of its coefficients

Exercise 2.B

The larger the sigma, the larger the mask.



σ=0.833 → 3σ = 2.5 → k=2



σ=2.5 → 3σ = 7.5 → k=7

See Stan Birchfield's chapter 5.2.4 on constructing Gaussian kernels

In this example, we apply another possible discretization criterion: $k = \lceil 3\sigma - 0.5 \rceil$

Exercise 2.B

You must create a function that:

- given σ, calculates the mask size (T), and provides a 1D Gaussian mask.
- given a mask size (T), automatically adjusts σ, and returns the 1D Gaussian mask.

T being the size of the mask, $T = 2 \cdot k + 1 \rightarrow k = (T-1)/2 \rightarrow (T-1)/2 = 3 \cdot \sigma \rightarrow \mathbf{2 \cdot [3 \cdot \sigma] + 1 = T}$

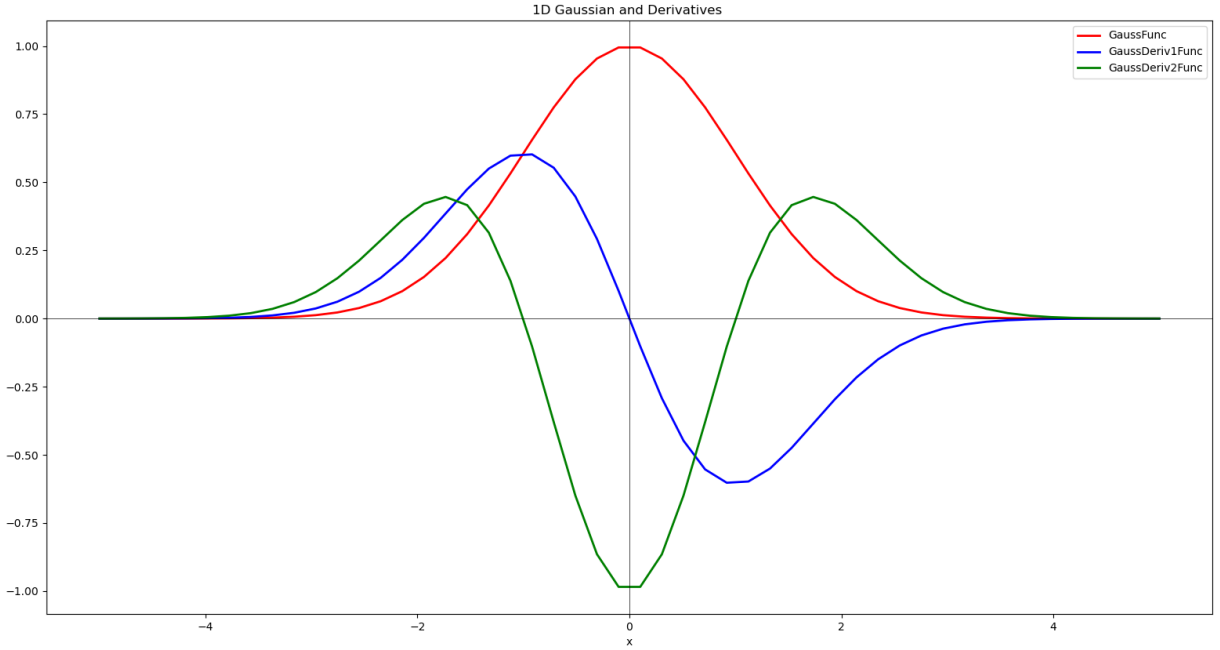
If given, for example, T=5:

• we can fill in the mask values by making [f(-2), f(-1), f(0), f(1), f(2)] and substituting in the Gaussian function a sigma of 0.67 (obtained from the previous formula).

Once we have σ and K we can discretize the mask by applying the Gaussian function (or its derivatives)

Exercise 2.B

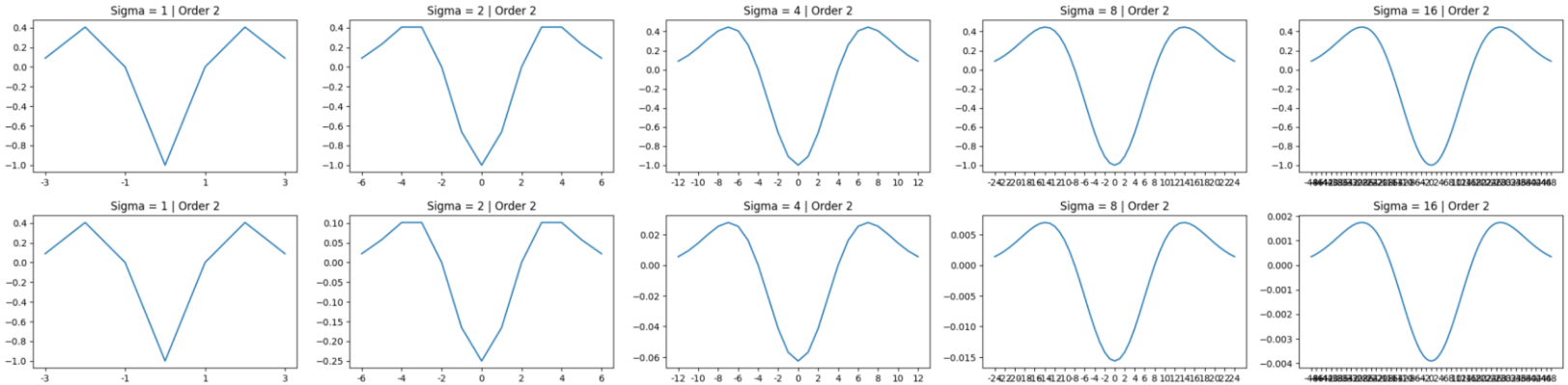
- The whole process described above applies to the derivatives of the Gaussian function



Exercise 2.B

How do we normalize the Gaussian derivative kernels? We normalize the scale!

To keep response the same (scale-invariant), must multiply Gaussian derivative by σ.
The second Gaussian derivative must be multiplied by σ^2.



Scale normalized

Scale NOT normalized

We'll look at this in more detail in Assignment 2 and during the theory sessions.

Exercise 2.B

- But… almost all the filters/kernels/masks we have seen were matrices (2D)…
  - Why are all these 1D vectors? 🤔
  - We will make use of the **separability of filters**: first, we convolve in one direction (1D) and then in the other (1D). We will come back to this later…

Exercise 2.B

- In the case of derivatives:
  - The only thing that changes is the function used to obtain the mask values (and the way we normalize the kernel).
    - You must calculate the derivatives and apply the resulting functions.
  - Remember that **the values of a derivative mask add up to approximately zero**.
    - **In the case of derivative masks, you should not divide them by the sum of the coefficients**
      - that would be 0, or a very small number → the kernel values would be huge.

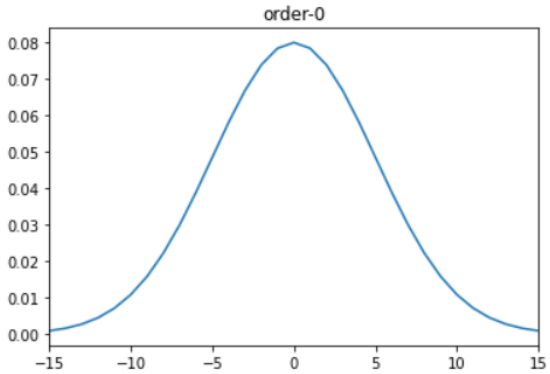
Why “approximately” and not “exactly” zero?

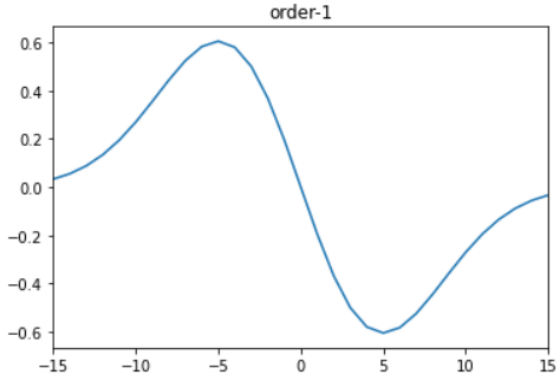


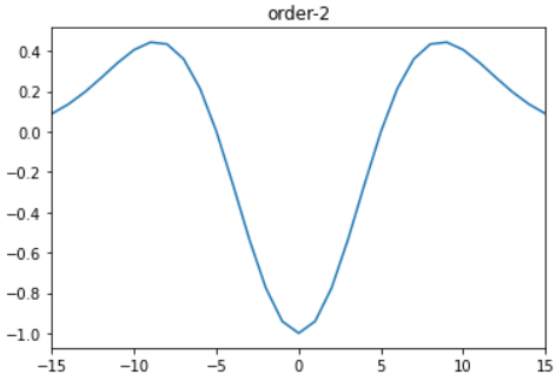
Exercise 2.B

- **Represent the profile (i.e. the silhouette of the mask as 1D functions) to verify that the masks created are correct**

Example of visual comparison of 1D masks







If you add the coefficients of the derivative kernels, which addition do you think will be closer to zero: the 1st derivative kernel or the 2nd derivative kernel?



How to present the exercises?

- The following tips apply to all exercises/sections.
  - https://pradogrado2627.ugr.es/pluginfile.php/153447/mod_folder/content/0/Algunas%20ideas%20sobre%20c%C3%B3mo%20elaborar%20un%20informe%20acad%C3%A9mico-cient%C3%ADfico.pdf
  - Check https://pradogrado2627.ugr.es/pluginfile.php/153448/mod_folder/content/0/Ejemplo_Otsu.ipynb
- Kernels should be visible and easy to analyze.
  - Adequate size to perceive nuances.
  - There is little point in putting them all in a single column, where the reader has to continuously scroll up/down (it makes it difficult to compare them)…
  - Figures with kernels must have a title to be easily identified.

How to present the exercises?

- The text accompanying the exercise must demonstrate your knowledge.
  - When only one or two short sentences are included as a comment (even if they are accurate) → impression of little effort.

- But… I don't know what to explain… ☹
  - Use your common sense! In Exercise 2.B, what does it seem logical to explain/show?
    - The analytical expressions of the calculated derivatives.
    - The relationship between σ and the mask size.
    - The importance of scale normalization: what is it and what is it used for.
    - Why derivative masks add up to zero and smoothing masks add up to one.
    - Analyze shape of kernels and why they have that particular shape.
    - …

Technical note: Float vs Int

```
filename = 'data/motorcycle.bmp'
im = read_image(filename,'gray',0).astype(np.float64)
GaussianKernel = kernelGauss1D(1,None,0)
image = insert_padding(im,GaussianKernel,2)

imageConvolved = convolve2D(image, GaussianKernel,
    GaussianKernel)
imGaussianBlur =
    cv.GaussianBlur(image,(len(GaussianKernel),len(G
    aussianKernel)),1,1)

compare_images(imageConvolved, imGaussianBlur)
```

Number of different pixels in both images (with a difference of more than 1 grey level): **1.081%** (1518 de 140454)

```
filename = 'data/motorcycle.bmp'
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    aussianKernel)),1,1)

compare_images(imageConvolved, imGaussianBlur)
```

Number of different pixels in both images (with a difference of more than 1 grey level): **8.117%** (11400 de 140454)

Check **`Guia_fundamentos_imagenes.ipynb`** to see, for instance, the difference between “cv2.add()” and “+” when dealing with np.uint8 values.

Final Remarks

- Remember to
  - Check the documentation:
    - https://docs.opencv.org/5.0
  - **Work on the discussion of results and explanation of the decisions taken, so that everything you do is clear and well justified**.
    - It's about making it easier for the reader to understand the work (and highlighting your efforts and knowledge)
      → **I cannot evaluate things that are not in the report!**

Final Remarks

- Figures must be easy to understand:
  - Appropriate resolution and size
  - Suitable title
  - They should be commented in a text cell (and, if they are not explicitly demanded in the exercise statement, their existence should be justified in some way)

Final Remarks

- About the discussion/explanation/analysis:
  - Check Ejemplo_Otsu.ipynb
  - It goes in text cells, not in code cells (as code comments do)
  - One should not
    - deviate from the point
    - be excessively brief
  - Do not hesitate to
    - link it to theory in the best way you can
    - incorporate additional experiments that you consider appropriate to demonstrate your hypotheses or to delve into each exercise
      - This is not necessary but, obviously, all the students' hard work and effort will be valued positively!
  - Avoid systematic spelling mistakes → they spoil the result!

Computer Vision

Assignment 1: Introduction to OpenCV, Responsible Use of Generative AI Tools, and First Steps in Image Processing

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